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Numerical investigation of hydrogen absorption in a metal hydride reactor with embedded embossed plate heat exchanger

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ABSTRACT

In this paper, a Metal Hydride (MH) reactor integrated with an Embossed Plate Heat Exchanger (EPHX) was studied for the first time for its hydrogen absorption rate and thermal performance. A detailed numerical analysis of the various flow-field designs in the EPHX such as parallel-type, pin-type, and serpentine types (vertical and horizontal) was performed. The serpentine flow-field EPHX presented better heat transfer and faster hydrogen storage ability. Also, it showed more uniform temperature distribution in the bed compared with parallel and pin-type flow-fields. Next, the vertical-serpentine flow-field EPHX was compared with the most commonly used Helical Coil Heat Exchanger (HCHX) and the outcomes were discussed. Although, EPHX showed slightly lower overall heat removal from the reactor, it presented similar hydrogen absorption rate and remarkable uniformity in temperature distribution in the reactor.

1. Introduction

In a Metal Hydride (MH) reactor, the heat transfer phenomena plays a significant role in regulating the time it takes to reach the equilibrium pressure of the hydride and hence the absorption rate. The heat transfer governs the temperature the hydride reach during the absorption, thus may have an influence on disproportionation. It is reported in a patent document [1], that cycling at high temperature can make the metal powder more likely to disproportionation. Many interesting strategies have been employed by various researchers to facilitate effective heat transfer from/to the reactor. In the last two decades, several investigators have proposed MH reactors integrated with different heat exchanging systems. These can be broadly classified as internal heat exchanger system, techniques to supplement effective thermal conductivity, and phase change materials. Some of the internal heat exchanger designs are spiral tubes [2,3], tube with fins [4–8], coiled tubes [9,10], helical coils [11–14], capillary tube bundles [15], double-layered annulus MH reactor [16–19], and heat pipes [20,21]. The techniques used to augment operative thermal conductivity of the reactor were metal wire matrix [22], multilayer sheet structure [23], MH powder composites [24], metal foam [25,26], and porous expanded graphite [27]. Several scholars devised MH reactors with phase change materials [28,29]. Hui et al. [30] developed a method to improve hydrogen absorption by physical moving of the powder by tilting the tank back and forth by 90° to enhance heat transfer rate. The results showed that larger MH reactors benefited more from the physical moving of the powder than the small sized reactors. A honeycomb structure heat exchanger with a resistive heater placed at the center was used by Claudio et al. [31] to study the desorption behavior of hydrogen in MOF-5® adsorbent material.

It is known that metal powder undergoes fragmentation into smaller particles upon repeated cycles of hydriding and dehydriding. This results in variation in the contact conditions between the alloy particles. A study by Kenta et al. [32] reveals the effect of pulverization on contact conditions between the alloy particles and hence its effective thermal conductivity. The major factors influencing the effective thermal conductivity of the particles are their contact ratio of point contact & plane contact, mean gap, and Biot number. After the charging and discharging cycles, although the increase in contact ratio of point contact increase the heat conduction, the significant effect of decrease in contact ratio of plane contact, mean gap and Biot number decrease the effective thermal conductivity of the alloy particles. The fine grains due to pulverization may gradually settle at lower portion of the cylinder under gravity forming a densely packed region. The denser zones are mainly due to segregation, ratcheting and followed by

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pulverization of the metal powder. Several research works have found that high-density regions may influence the rate of hydrogenation and produce a great amount of strain, which could deform/rupture the reactor, in particular thin-wall reactors. Sidharth and Mohan [33] studied wall strain development upon hydrogenation, and also the influence of aluminum foam and its density on wall strain. Several studies in literature have focused on the influence of pulverization, redistribution, and agglomeration of metal powder during cyclic hydriding and dehydriding reactions on stress development in the reactor [34–37]. Charlas et al. [38] examined the cyclic swelling of the intermetallic BCC compound T_i-C_r-V + Z_r-N_i in one granular bed and highlighted the gradual settlement of the hydride with continuous reduction in grain size due to decrepitation during successive cycles. The influence of non-uniform heat removal/supply (temperature gradients) and hence non-uniform thermal cycling in the reactor on pulverization process & densification is not investigated to date. However, in an ideal condition, it is interesting to have uniform heat removal i.e. homogeneous temperature field and hence uniform utilization of metal powder in the reactor. Chippar et al. [39] proposed a stack reactor configuration for storage of hydrogen and showed that compartmentalization of the reactor can enhance the hydrogen absorption rate due to better heat removal, and also improve uniformity in temperature distribution.

Embossed Plate Heat Exchangers (EPHX), one of the popular types of HXs, are compact in size, lightweight, and have high heat transfer capacity. They are widely used in many industrial applications such as food and chemical engineering, HVAC, automobile radiator, refrigeration, aerospace, and other engineering areas. Based on the use, different types of channels: plain, wavy, offset strip, perforated, and louvered are used in the EPHX. Comparative studies [40–43] of the various types of channels in the EPHX indicate that vortex generators are very effective in heat transfer improvement. Vortex generators are tiny interrupting surfaces of rectangular, delta or any other shape, which can be punched, mounted, attached, embossed or stamped in the boundary layer. They not only disturb the flow, but also generate intensive vortices and generate secondary flows, which improve the convective heat transfer rate. Numerous experimental and numerical studies have been conducted on thermal-hydraulic characterization of vortex generators in the channels. However, literature shows no studies so far on using EPHX in the MH reactors.

In the present study, the heat transfer in an MH reactor has been explored by introducing a single EPHX. The numerical model used in this study is similar to the several models available in the literature, however the original contribution of this article is proposal of EPHX for MH reactor for the first time. Several flow-field configurations were employed such as parallel type, pin-type, and serpentine types (vertical and horizontal) in the plate, and their significance was studied on heat transfer rate, hydrogen storage, and temperature distribution in the reactor. Finally, the thermal performance of the vertical-serpentine flow-field EPHX and the Helical Coil Heat Exchanger (HCHX) was compared.

2. Mathematical model

2.1. Conservation equations

The present model is built on a previous work [39]. The mass, momentum, and energy conservation equations were solved in this model along with various constitutive relations. The readers are requested to refer to the previous publication for a more detailed description of the model.

The mass conservation of hydrogen gas writes:

$$\frac{\partial \varepsilon \rho^g}{\partial t} + \nabla \cdot (\rho^g \vec{u}) = -\dot{m}. \tag{1}$$

Hydrogen is assumed to be an ideal gas and hence,

$$\rho^g = \frac{P^g M^g}{RT} \tag{2}$$

The mass conservation of metal writes:

$$\frac{(1-\varepsilon)\partial \rho^s}{\partial t} = \dot{m}. \tag{3}$$

where $\dot{m}$ represents the hydrogen absorption rate per unit volume and is determined by the reaction kinetics of $LaNi_5$ as [44]:

$$\dot{m} = C_a \exp\left(-\frac{E_a}{RT}\right) \cdot \ln\left(\frac{P^g}{P_{eq}}\right)\left(\rho^s_{sat} - \rho^s\right) \tag{4}$$

The equilibrium pressure, P_{eq} in the MH bed is obtained as a function of temperature and hydrogen to metal (H/M) ratio as [39,45]:

$$P_{eq}(\text{bar}) = f\left(\frac{H}{M}\right) \cdot \exp\left(\frac{\Delta H}{R_g}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right) \tag{5}$$

The momentum conservation of hydrogen flow is written as follows:

$$\frac{1}{\varepsilon}\frac{\partial \rho^g \vec{u}}{\partial t} + \frac{1}{\varepsilon^2}\nabla \cdot (\rho^g \vec{u}\vec{u}) = -\nabla p + \nabla \cdot \tau + S_u \tag{6}$$

The momentum source term in the metal bed was calculated using Darcy's law - $S_u = -\frac{\mu}{K}\vec{u}$, while in the other regions -$S_u = 0$.

Energy conservation:

The temperature field in the porous and non-porous regions was solved using the standard equations provided by ANSYS-FLUENT as follows:

Non-porous medium

$$\frac{\partial(\rho^g E^g)}{\partial t} + \nabla \cdot \left(\vec{u}\left(\rho^g E^g + P\right)\right) = -\nabla\left(k^g \nabla T + \tau.\vec{u}\right) \tag{7}$$

Porous medium

$$\frac{\partial\left(\varepsilon \rho^g E^g + (1-\varepsilon)\rho^s E^s\right)}{\partial t} + \nabla \cdot \left(\vec{u}\left(\rho^g E^g + P\right)\right) = -\nabla\left[k_{eff}\nabla T + \tau.\vec{u}\right] - \dot{m}\left[\Delta H + T\left(C^g_P - C^s_P\right)\right] \tag{8}$$

Table 1
Operating conditions and thermo-physical properties.

Description	Value	Ref.
Initial & inlet temperature, T_0&T_{in}	293.15 K	[45]
Inlet pressure, P_{in}	10 atm	[45]
Reference temperature, T_{ref}	293.15 K	[45]
Activation energy, E_a	21,179.6 J mol^{-1} K^{-1}	[46]
Heat of formation, ΔH	-1.54×10^7 J kg^{-1}	[46]
Absorption rate coefficient, C_a	59.187 s^{-1}	[46]
Specific heat of hydrogen, C_{P,H_2}	14,890 J mol^{-1} K^{-1}	[46]
Specific heat of metal, $C_{P,M}$	419 J mol^{-1} K^{-1}	[46]
Thermal conductivity of hydrogen, κ_{H_2}	0.167 W m^{-1} K^{-1}	[27]
Thermal conductivity of metal, κ_M	1.5 W m^{-1} K^{-1}	[45]
Porosity of metal bed, ε	0.63	[45]
Hydrogen-free metal density, ρ^s_0	8175 kg m^{-3}	[22]
Permeability of the metal bed, K	10^{-8} m^2	[22]
Thermal conductivity of stainless-steel plates	16.26 W m^{-1} K^{-1}	

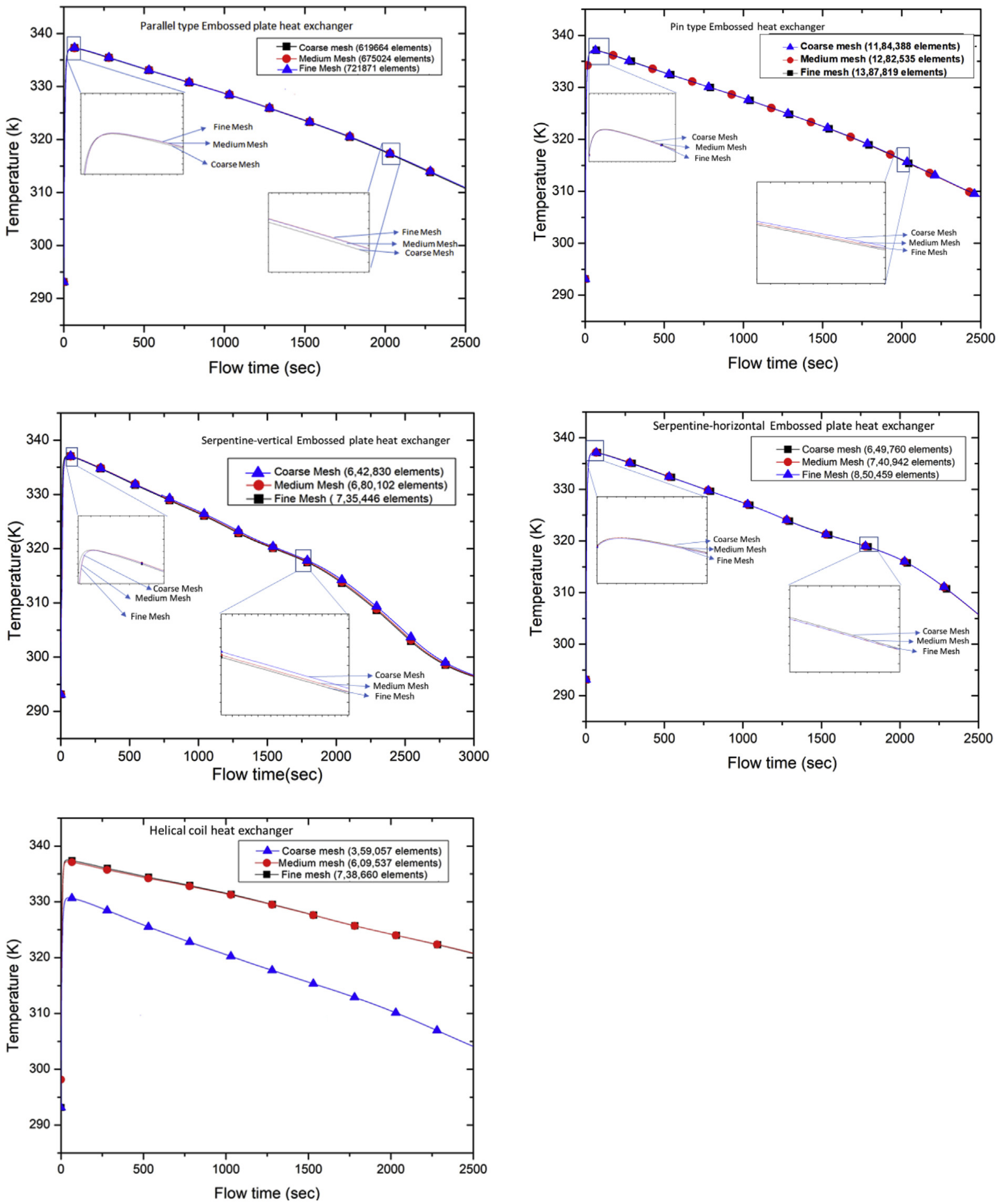

Fig. 1. Grid independency analyses for parallel, pin-type, vertical serpentine, horizontal serpentine EPHX and helical coil heat exchanger.

The effective thermal conductivity in the bed was determined as a function of porosity-weighted function of hydrogen gas, and solid medium thermal conductivities as below:

$$k_{\mathrm{eff}} = \varepsilon k_{H_2} + (1-\varepsilon)k_M \tag{9}$$

2.2. Initial/boundary conditions and numerical implementation

Initially, a uniform metal density, i.e., equal to empty metal density, temperature, and pressure was defined. At the inlet of the hydrogen gas, a constant supply pressure and temperature were defined. No-flux conditions were applied to the surrounding wall of the reactor. Table 1 should be referred to for the various operating conditions and thermo-physical properties of the metal powder used in the simulations.

$$\rho^s = \rho_0^s;\ T = T_0; P^g = P_0;\ P_{in} = 10\ \text{atm}; T_{in} = 293.15\ \text{K} \tag{10}$$

The exothermic heat due to hydrogen absorption was removed by the water supplied in the coolant channel, and the coolant flow rate and temperature at the inlet were 0.0004639 kg/s and 293.15 K, respectively. The coolant flow rate is approximated as a function of total amount of heat generated during the absorption and the coolant temperature rise (taken as 10 °C) from inlet to outlet in the heat exchanger. The above unsteady model was numerically implemented into a commercial Computational Fluid Dynamics (CFD) software package, ANSYS-FLUENT 17.2 based on the finite volume method. The source terms in the governing equations and other constitutive equations were written as User Defined Function (UDF) and compiled in to the solver. The convergence criteria for the mass, momentum, and energy

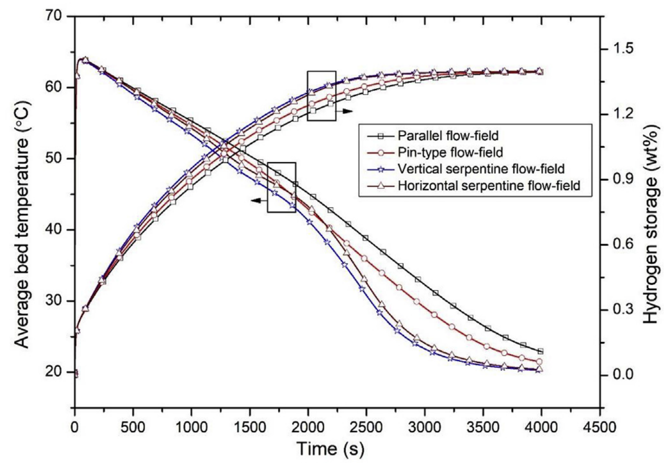

Fig. 3. Average bed temperature and hydrogen storage evolution curves in the reactor with parallel, pin-type, vertical serpentine and horizontal serpentine EPHX.

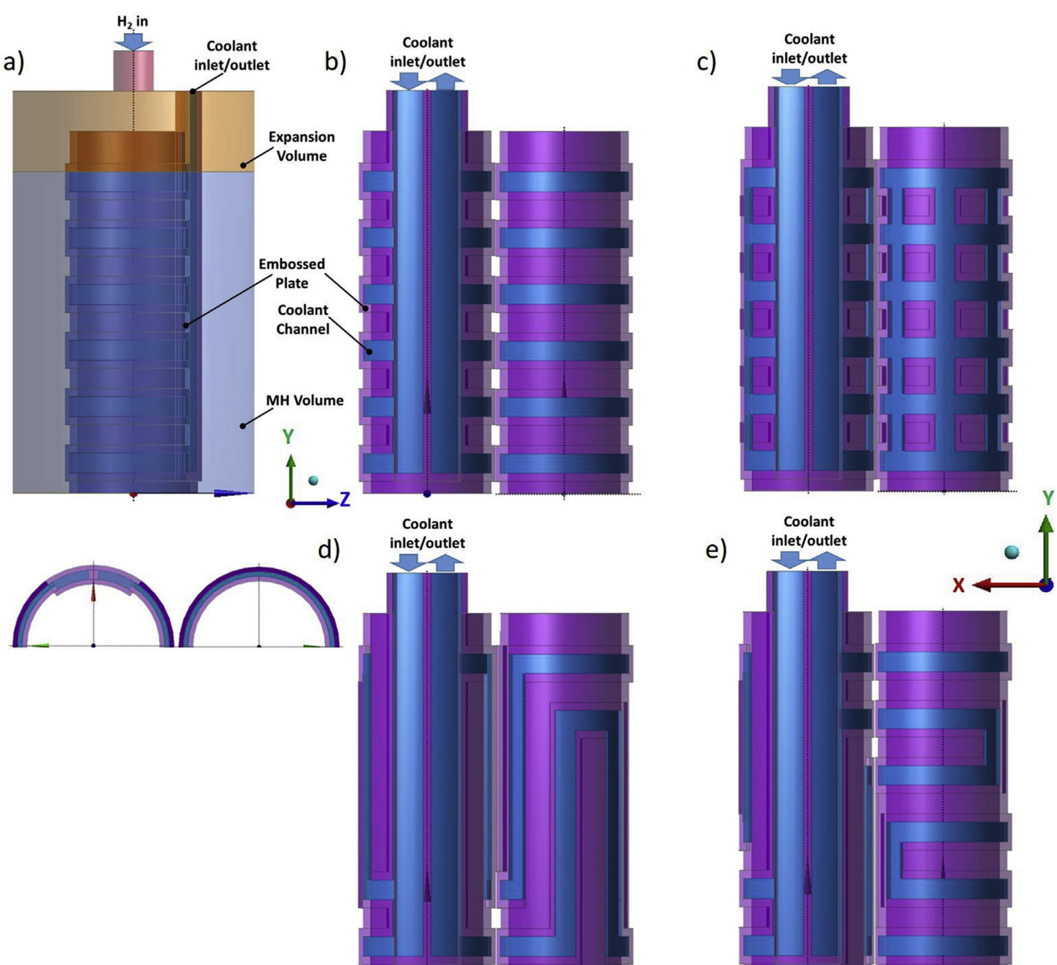

Fig. 2. Flow-field configurations of EPHX: a) MH reactor with parallel flow-field EPHX; (b), (c), (d), and (e) longitudinal section depicting the two halves of the parallel, pin-type, vertical serpentine, and horizontal serpentine EPHX.

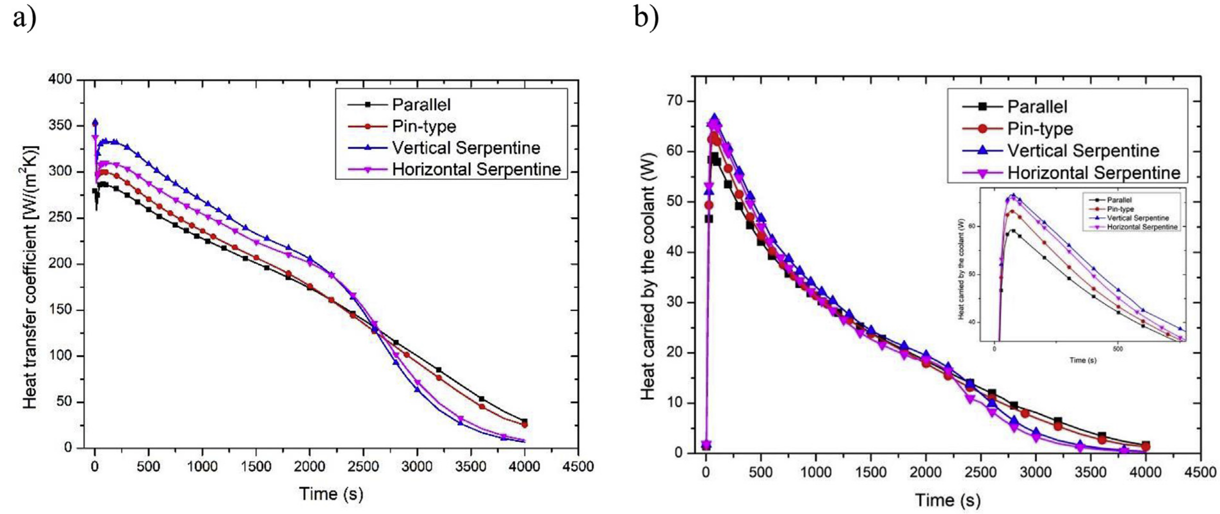

Fig. 4. Heat transfer coefficient and heat carried away by the coolant with parallel, pin-type, vertical serpentine and horizontal serpentine EPHX.

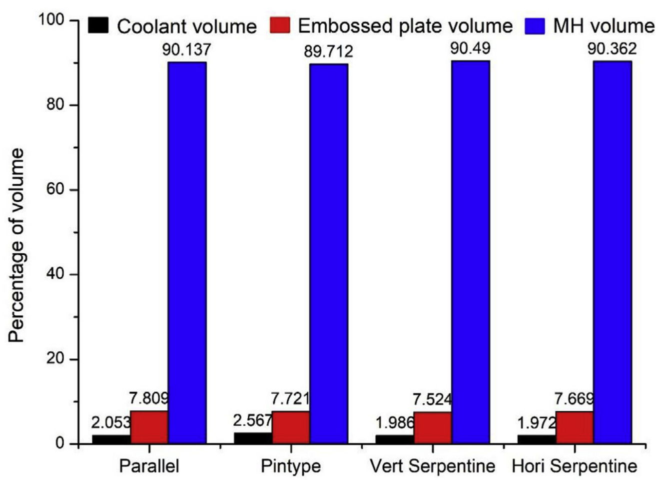

Fig. 5. of embossed plate, coolant flow-field, and metal bed volumes in the reactor with different types of flow-field layouts in EPHX.

calculation residuals were set at10-6. A detailed grid-independence tests were performed for various flow-field designs by testing the effect of grid resolution on the evolution of average bed temperature. The grid-independence test results are presented in Fig. 1. As shown in Fig. 1a, for parallel flow-field, the temperature curves slightly differ with coarse and medium grid resolution, however, no difference in result was obtained between medium and fine meshes, and hence medium mesh was chosen for further simulations. Similar analysis were performed for the remaining flow-field designs.

3. Results and discussion

A cylindrical tank of 60 mm diameter and 100 mm height was used, in which $LaNi_5$ powder was filled to a height of 80 mm. An EPHX of two stainless-steel plates of 2 mm thickness each was prepared. The first plate had a stamped/embossed flow-field pattern and was joined to the second plate. The plates can be joined by using laser or resistance welding. The HX was placed coaxial and fixed to the bottom of the tank. The height and mean

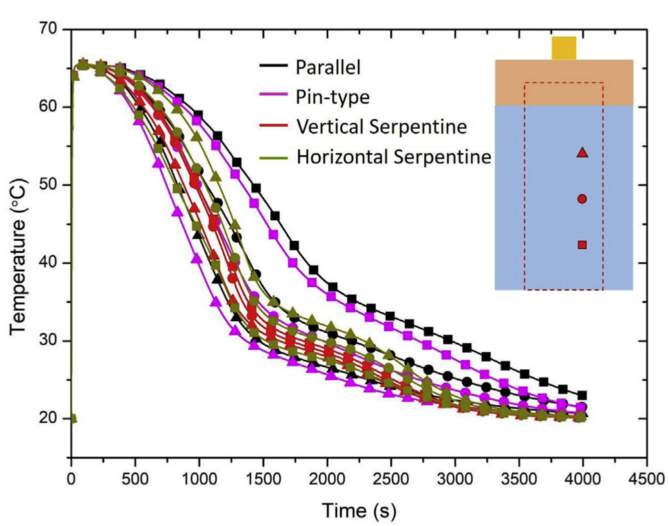

Fig. 6. Temperatures monitored at three different locations at the inner side of the EPHXs.

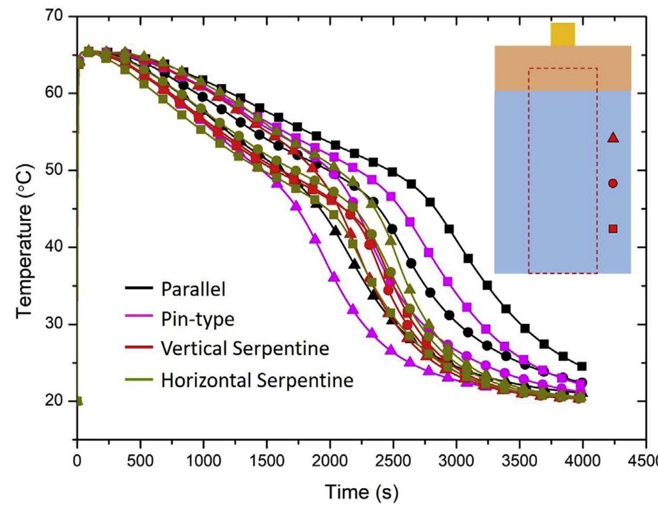

Fig. 7. Temperatures monitored at three different locations at the outer side of the EPHXs.

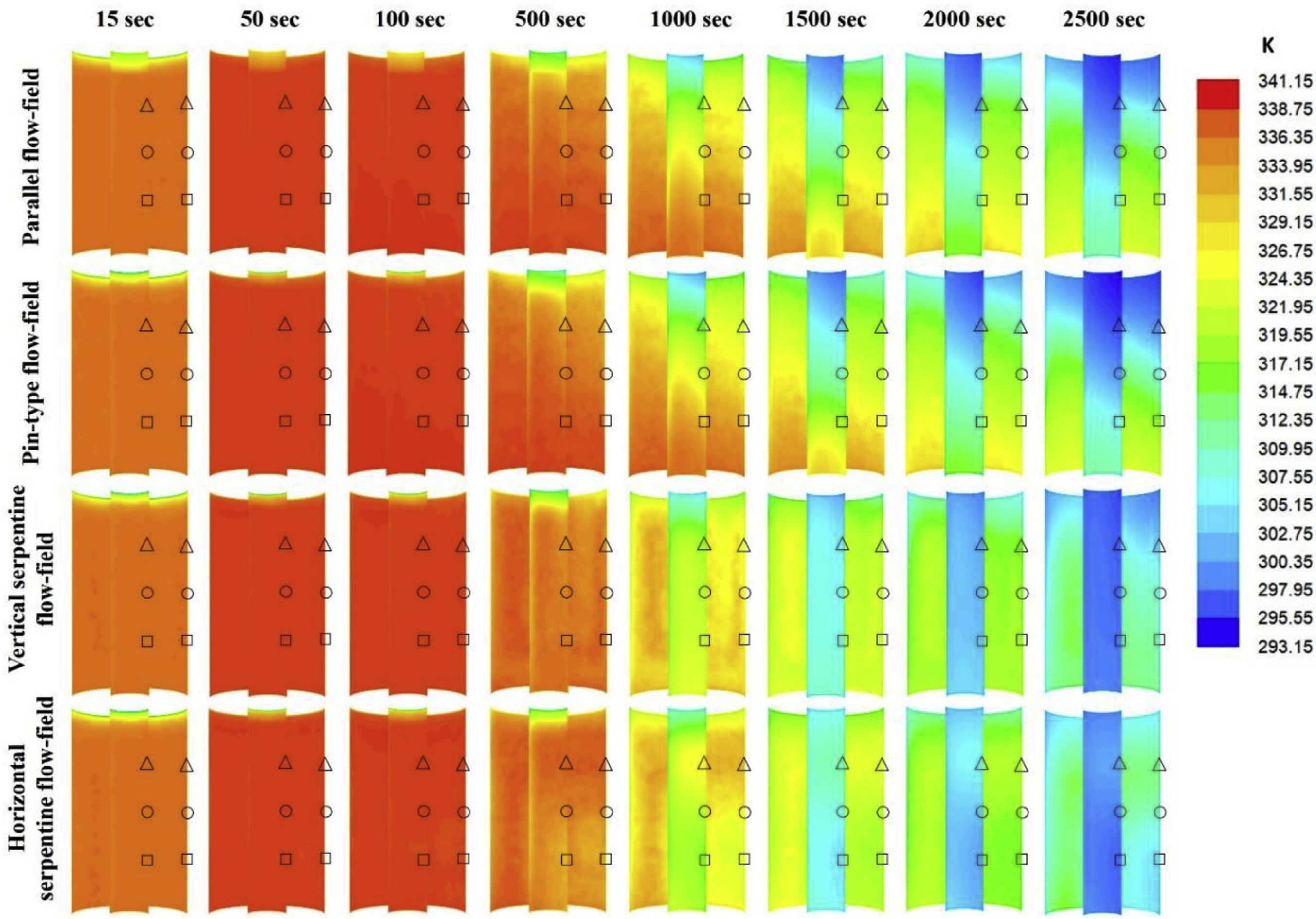

Fig. 8. Contours of local temperature at inner (R = 7.5 mm) and outer (R = 22.5 mm) surface of the EPHX (▭ h = 20 mm, ◯ h = 40 mm, Δ h = 60 mm).

diameter of the HX were 85 mm and 30 mm, respectively. Various flow-field patterns such as parallel, pin-type, and serpentine (horizontal and vertical) were used in the study. The embossing height, width, and pitch were 2 mm, 5 mm, and 10 mm, respectively, for all the flow-field configurations. The inlet and outlet manifolds were created by embossing both the first and second plates, and the size was 10 mm (width) x 4 mm (height). Fig. 2 shows the various flow-field patterns used in the present study. Fig. 2 (a) shows the MH reactor with parallel flow-field EPHX and Fig. 2(b)–(e) show the longitudinal section depicting the two halves of the parallel, pin-type, vertical serpentine, and horizontal serpentine EPHX, respectively. The pin-type flow-field has projections called as pins, and the major characteristics are formation of stagnant areas, low pressure drop and non-uniform distribution of the coolant as it flows through the least resistance path. In parallel flow-field, the plate is divided into symmetric flow sectors, having alternative channels and lands. Each flow sector has separate inlet and outlet connected to the supply and exit manifolds. The serpentine flow-field design has single (or multiple path) continuous fluid flow channels from supply to exit manifold, resulting in high pressure drop.

Fig. 3 shows the average temperature and hydrogen storage (wt.%) in the MH reactor with various flow-field layouts in the EPHX. Comparing the temperature curves, it was obvious that the serpentine channels showed remarkable improvement in heat removal out of the reactor than the straight (parallel and pin-type) channels. This was justified by invoking the Nusselt number. The heat transfer between the solid surface and adjacent fluid is directly proportional to the Nusselt number and is determined by both the shape and route of the channel path. The Nusselt number for the parallel channels will stay constant as long as the flow conditions stay constant and fully developed. The higher thermal performance of the serpentine channels could be observed because of the tortuous path of the fluid. The pressure drops in the parallel, pin-type, vertical serpentine, and horizontal serpentine channels were 32.8 Pa, 31.6 Pa, 209.6 Pa, and 194.1 Pa, respectively. Among the serpentine flow-fields, the vertically laid channels showed slightly better heat transfer ability than the horizontal one. In the vertical serpentine flow-field, the coolant flow periodically between the top and bottom portion of the reactor resulted in more uniform and fast heat removal from the bed. This will be later discussed in detail. Corresponding to their thermal performance, the 90% absorption saturation time for MH reactors with parallel, pin-type, vertical serpentine, and horizontal serpentine EPHX was 2236 s, 2092 s, 1863 s, and 1897 s, respectively. Fig. 4 shows the heat transfer coefficient and heat carried away by the coolant in various flow-field designs. The increase in the heat transfer coefficient with vertical serpentine flow-field is around 19%, 14% and 7% more than the parallel, pin-type and horizontal flow-fields, respectively up to 2500 s. Similarly, heat removed with the vertical serpentine flow-

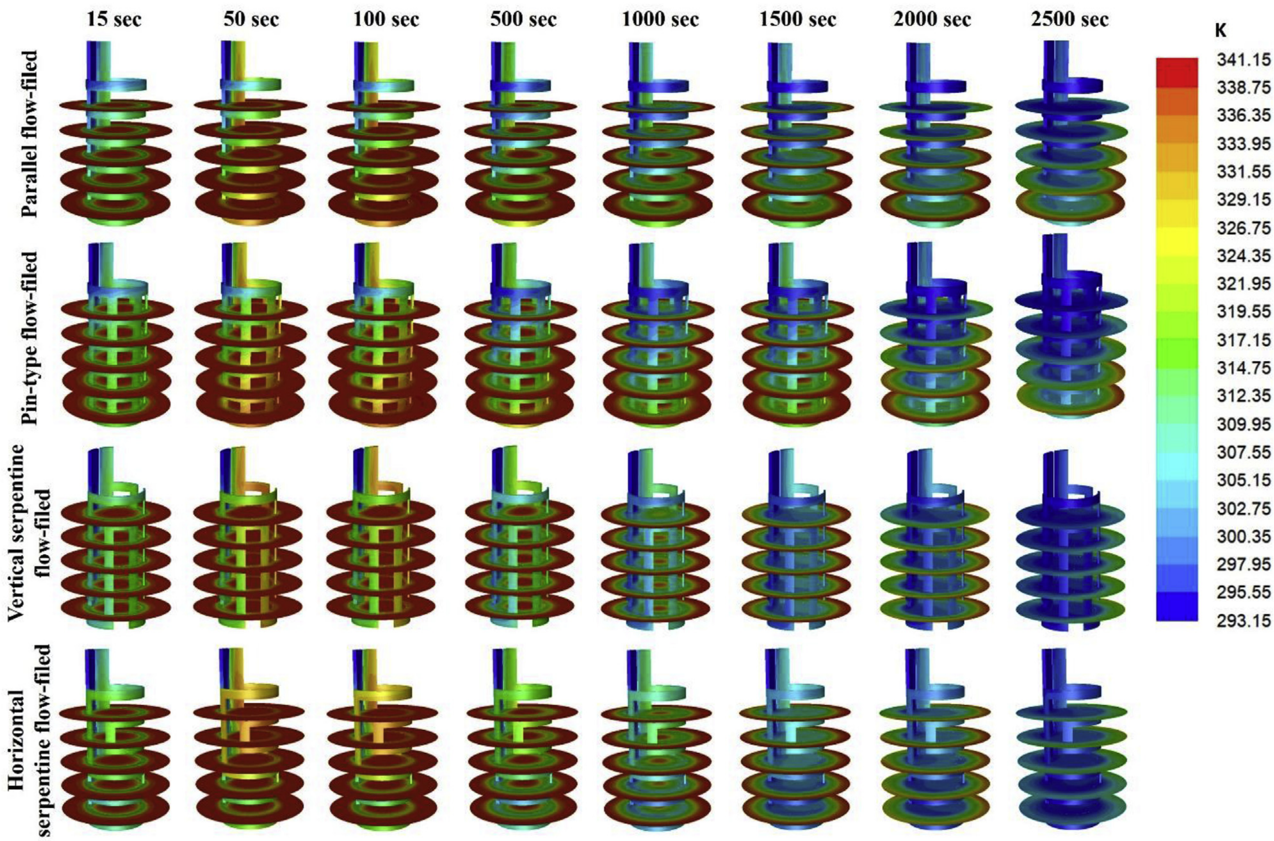

Fig. 9. Contours of local temperature for EPHXs with different flow-field configurations.

field is higher than the other flow-field designs. The hydrogen absorption process for horizontal and vertical serpentine flow-fields retard around 2500 s, and concurrently, the heat transfer coefficient and heat carried away by the coolant decreases at a faster rate.

It is important to achieve a balance between the HX volume and the volume available for the metal powder in the reactor. For a realistic comparison, the volumes occupied by the embossed plates with different flow-field configurations were maintained nearly same. Fig. 5 shows the percentage of embossed plate, coolant flow-field volume, and actual volume available for the metal powder in the reactor. To analyze the distribution of temperature within the reactor, it was monitored at six locations, three points each inside and outside of the EPHX (h = 20 mm, 40 mm and 60 mm) at a radius of 7.5 mm and 22.5 mm, respectively. The temperature evolution at the monitored locations was plotted in Figs. 4 and 5 at the inner and outer sides of the EPHX, respectively. From Fig. 6, it is clear that the difference in temperature evolution at the three monitored points with parallel and pin-type EPHX was large, indicating non-uniform heat removal/temperature distribution. With the serpentine flow-field EPHX, the temperature gap between the monitored points was reduced and in particular, highly uniform temperature distribution was observed with the vertical serpentine EPHX. For instance, at 75% saturation of the metal powder, the temperature difference between the bottom and top points for parallel, pin-type, vertical serpentine, and horizontal serpentine EPHX were 19.5 °C, 19.3 °C, 4.8 °C, and 10.78 °C, respectively. A similar trend was observed at the monitoring points located outside of the EPHX as shown in Fig. 7. For a better assessment of longitudinal gradient, a multidimensional temperature distribution in inner and outer surfaces of the MH bed at various time intervals is presented in Fig. 8. As the absorption proceeds, parallel and pin-type flow-field EPHXs show large temperature difference, whereas a small and constant difference is observed with the serpentine flow-field EPHXs. Comparing Figs. 6 and 7, a temperature gradient of 16 °C is observed in the radial direction (between the line of points inside and outside of the embossed plates) as the radial gradient is governed by the hydride powder thermal conductivity alone. It should be noted that the radial position chosen for the EPHX (i.e. R = 15 mm) may not be the optimum to equilibrate the reacted fraction inside and outside. In a large reactor, a big gradient in temperature in the radial direction can be reduced by stacking several concentric EPHX.

Fig. 9 shows the temperature distribution at five fractional locations along the axis of the reactor and the coolant channel for various flow-fields at different time intervals. In all cases, the bed temperature increases sharply during the initial stages due to fast hydrogen absorption in the bed. The difference in temperature distribution with different flow-fields can be observed around 500 s and as the absorption proceeds, the EPHX with serpentine flow-fields shows superior performance compared with the parallel and pin-type flow-field EPHXs. A significant disparity in temperature distribution in the bed is clearly visible even after 2500 s for parallel and pin-type flow-fields. These results reveal the importance of coolant flow direction in the bed, i.e., both radial and longitudinal directions in the serpentine flow-field, whereas, mainly in radial direction in straight flow-fields (parallel and pin-

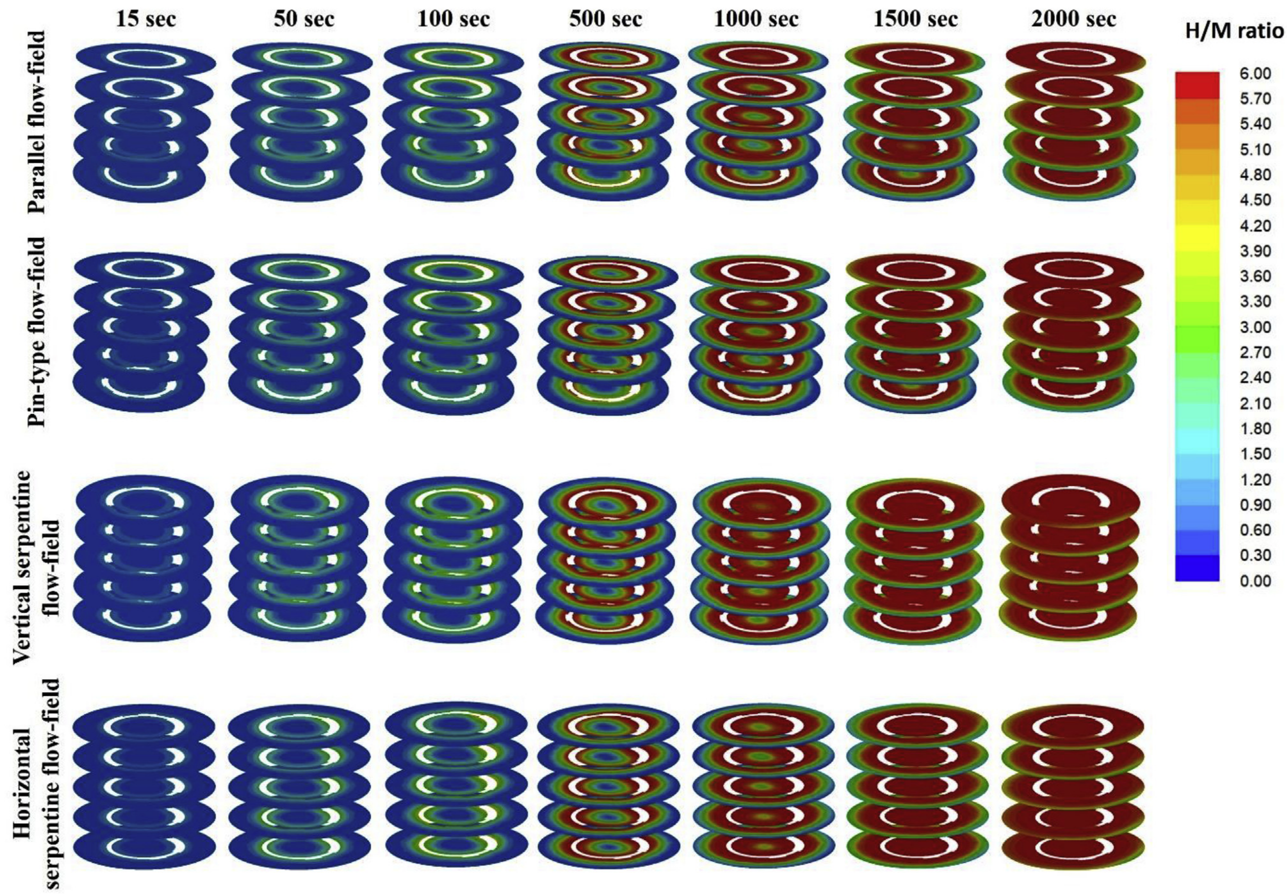

Fig. 10. Contours of local H/M ratio for EPHXs with different flow-field configurations.

type). Fig. 10 shows the H/M ratio distribution in the metal bed. As expected, hydrogen was absorbed rapidly near the EPHX due to greater heat transfer compared with the other regions. Owing to better heat transfer, the EPHX with serpentine channels showed uniform hydrogen absorption compared with the two other types of flow-fields.

The effectiveness of the EPHX in the reactor was demonstrated numerically by comparing the thermal performance and absorption rate with a reactor incorporated with the more widely used HCHX. The volume occupied by the HCHX in the reactor and coolant inlet boundary condition were maintained similar to that of the EPHX. The outer diameter, inner diameter, and the number of turns of the tube are 6 mm, 4.8 mm, and 8, respectively. Fig. 11 compares the average bed temperature, hydrogen storage, and temperature monitored at six locations for the vertical serpentine EPHX and HCHX. From Fig. 11a, it was realized that the average bed temperature was low for the HCHX indicating higher overall heat removal from the reactor. However, the hydrogen absorption rate for HCHX is same as that of the EPHX. These results indicate a significant spatial variation in heat removal with the HCHX. Fig. 11 (b and c) showing temperature evolutions at three locations each along the axis inside and outside of the HX supports this result. With HCHX, the temperature significantly increases along the direction of the coolant flow presenting the hottest area near the bottom of the reactor, whereas, the temperature variation was minimum with the EPHX. Similar to Figs. 8 and 12 show the local temperature distribution in inner and outer surfaces of the EPHX and HCHX in the MH bed at various intervals of time. It is clear from the contours that the EPHX presents a small longitudinal temperature gradient than the HCHX.

Spatial variations in temperature and absorption rate are well-presented with multi-dimensional contours in Fig. 13. From Fig. 13a, it is clear that non-uniformity in temperature distribution significantly increases along the direction of the coolant flow, i.e., from the top to the bottom of the reactor with HCHX, whereas great uniformity is observed with EPHX. This is mainly due to the direction of the coolant flow. In the case of EPHX, there is always periodical coolant flow from hot zone (bottom of the reactor) to cold zone (top of the reactor), helping to maintain uniform temperature in the reactor. Temperature variation along the radial direction is the same in both the cases. Following temperature distribution, the hydrogen absorption was highly non-uniform for HCHX and very uniform for EPHX.

The convective heat transfer coefficient in the embossed channels can be enhanced by several mechanisms that include promoting swirl or vortex flows, small hydraulic diameter of the channel, disruption and reattachment of boundary layers, etc. This will be a subject for future studies.

4. Conclusions

The compact size and high heat transfer capacity of the EPHX makes them very competitive compared with other types of HXs. The present work aimed at exploring the use of EPHX in a cylindrical MH reactor. It used different flow-field layouts in the EPHX such as parallel, pin-type, and serpentine (vertical and horizontal). A detailed numerical study was performed to investigate the influence of flow-field layouts on heat transfer and its concomitant

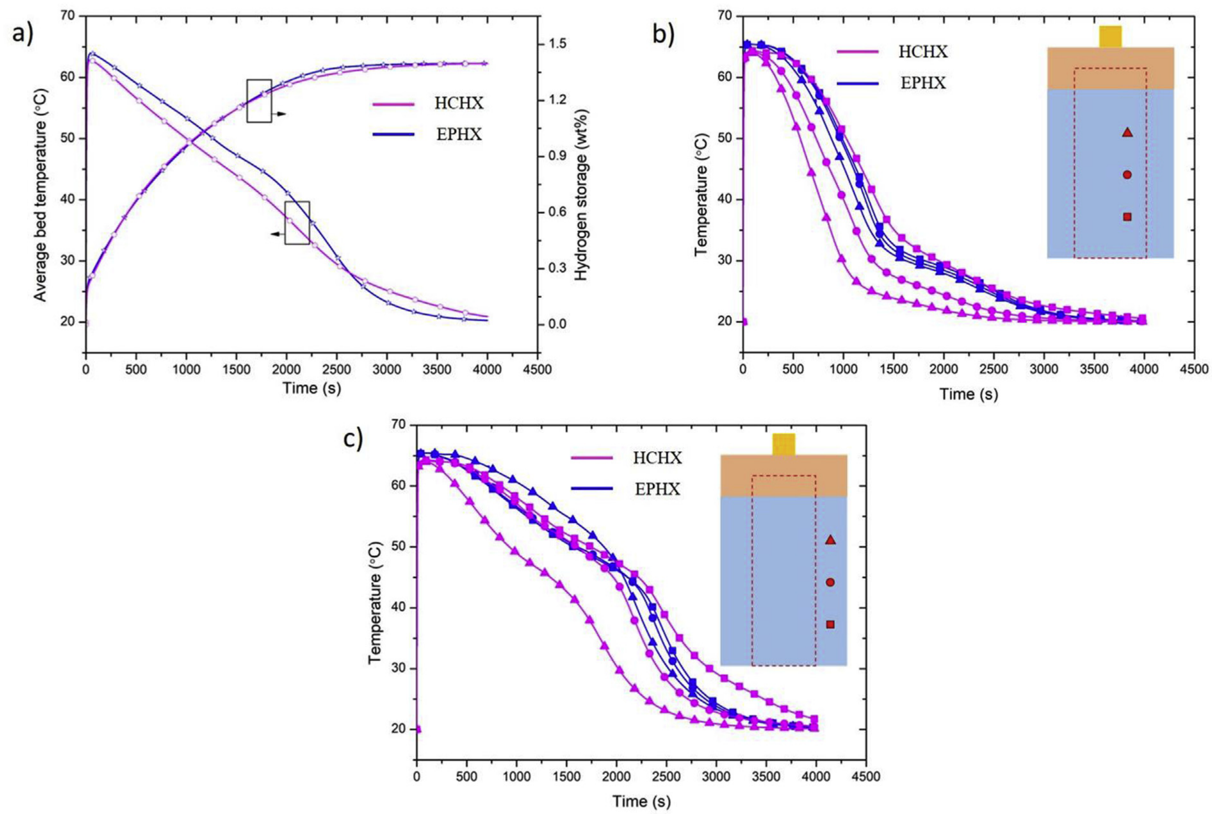

Fig. 11. Comparison of average bed temperature, hydrogen storage and temperatures monitored at six locations in the MH reactor with HCHX and EPHX, separately.

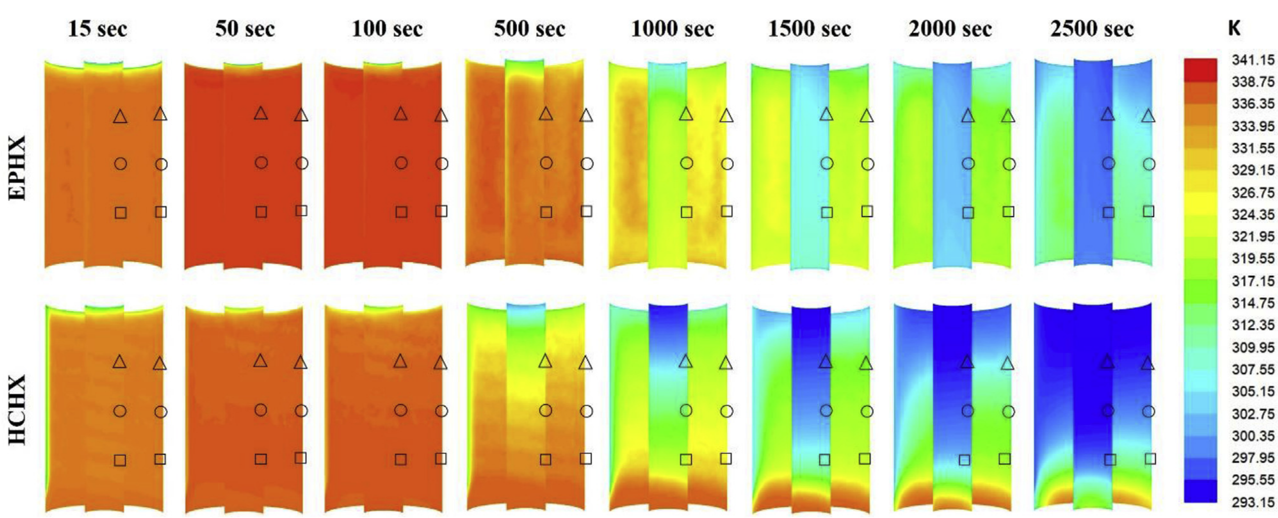

Fig. 12. Contours of local temperature at inner (R = 7.5 mm) and outer (R = 22.5 mm) surface of the EPHX and HCHX (▭ h = 20 mm, ○ h = 40 mm, Δ h = 60 mm).

effect on the hydrogen absorption process. A comparison of thermal performance and hydrogen absorption in the bed showed that the serpentine layouts exhibited much better performance than the parallel and pin-type layouts. Further studies were carried out to assess the influence of flow-field layouts on temperature distribution in the bed. Along the radial direction, all the layouts exhibited equal variation in temperature, but the vertical serpentine channels showed significantly smaller variation in longitudinal direction. Further, a detailed multidimensional contour of temperature and H/M ratio distribution in the MH reactor were

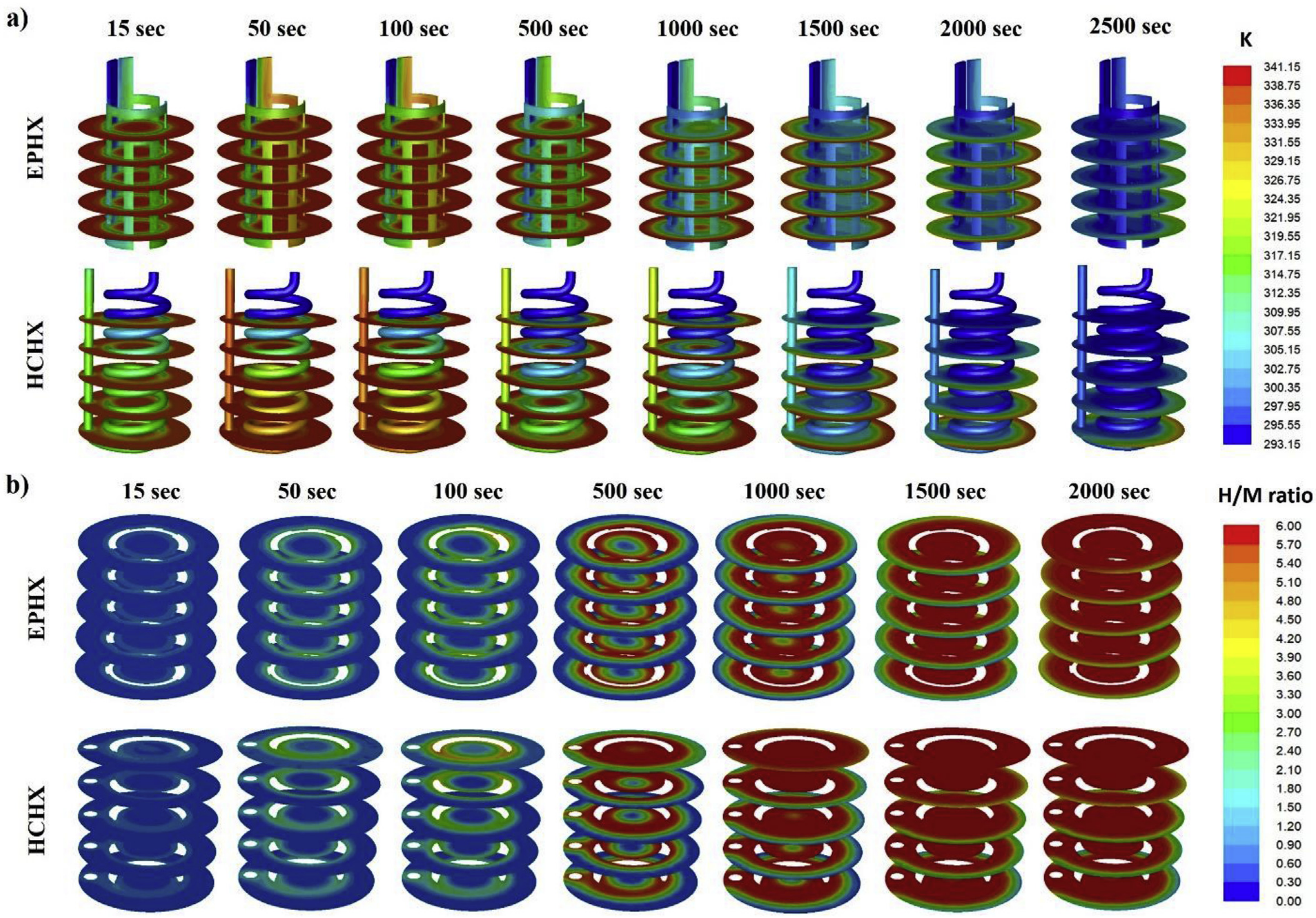

Fig. 13. Contours of local (a) temperature and (b) H/M ratio in the MH reactor with EPHX and HCHX.

presented. Finally, the thermal performance and hydrogen absorption rate of the vertical serpentine EPHX and HCHX were compared. The results showed that the EPHX helps in maintaining a uniform temperature distribution and eventually, uniform metal powder utilization in the reactor.

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Nomenclature

C_a: absorption rate constant, s^{-1}
C_P: specific heat, kJ.(kg K) $^{-1}$
E_a: activation Energy, J.mol^{-1}
$\frac{H}{M}$: hydrogen to metal atomic ratio
ΔH: reaction heat of formation, J.kg^{-1}
h: heat transfer coefficient, W.(m^{2}K)$^{-1}$
K: permeability, m^2
M: molecular weight, kg.mol^{-1}
P: permeability, m^2
R: universal gas constant, 8.314 J (mol K)$^{-1}$
T: temperature, K
$\vec{u}$: velocity vector. m.s^{-1}
E: total energy J/kg
k: Thermal conductivity, W.(m K) $^{-1}$

Greeks

ε: porosity
ρ: density, kg.m^{-3}
k: thermal conductivity, W.(mK)$^{-1}$
μ: dynamic viscosity, kg.(m s)$^{-1}$

Superscripts

g: gas phase
s: solid phase
sat: saturation value

Subscripts

eq: Equilibrium
H_2: Hydrogen
M: Metal
ref: reference value
0: initial value
eff: Effective